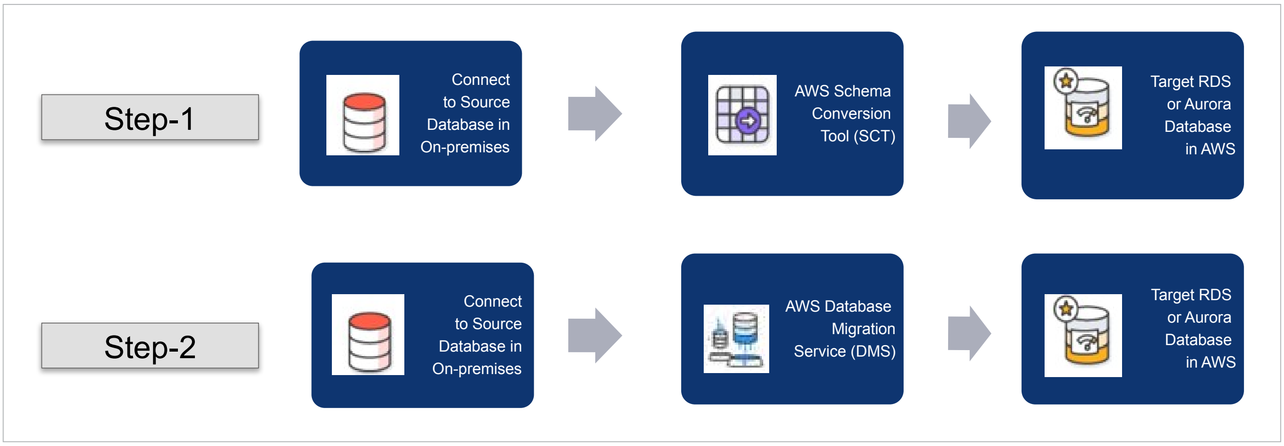

Figure 1: Steps in AWS native database migration

Tool (SCT) for migrating data objects and structures from source to target, encompassing tables, indexes, data constraints, functions, and procedures, among others. Once completed, the target database in AWS will have the like-to-like structure of the source, enabling the use of AWS data migration service (DMS) tools. Figure 1 shows these steps in order.

DMS ensures high reliability and recovery of data migration, supporting scheduled migrations to migrate data during weekends, minimising live data traffic in databases from running applications. Also, using S3 bucket for storing buffered data expedites migration to AWS database, and DirectConnect facilitates connection and transfer of data from on-premise to the cloud. For higher data security, DMS encrypts data in transit during migration ensuring data integrity.

Architectural decisions on choosing the right database solution for the target architecture hinge on various factors, including functional and non-functional requirements (NFR). Table 1 lists various AWS database services and their potential use cases aiding in decision-making for selecting the right database solutions.

Right-sizing database solutions in AWS

Right-sizing is the process of matching the database resources with your actual workload requirements. This involves analysing your database usage patterns, such as CPU, memory, and I/O, and then selecting the instance type and size that offers the best performance for your workload at the lowest possible cost.

Right-sizing your database solutions in AWS is an essential step in optimising your cloud costs and ensuring that your databases are at peak performance. By selecting the right instance type and size for your workload, you can avoid both overprovisioning and under-provisioning, mitigating unnecessary costs and potential performance bottlenecks.

Right-sizing is important for several reasons.

Cost *optimisation:* By right-sizing your databases, you avoid paying for unused resources, resulting in significant cost savings on your AWS bill.

***Performance optimisation**:* Under-provisioned databases may struggle to keep up with workloads, leading to performance bottlenecks. Conversely, overprovisioned databases incur costs for unused resources.

***Compliance**:* Certain regulatory requirements mandate that your databases be right-sized. For example, the HIPAA regulations require that healthcare organisations right-size their databases to protect patient data.

How to right-size your databases in AWS

There are several steps you can take to right-size your databases in AWS.

***Analyse your database usage patterns**:* Utilise AWS CloudWatch or other monitoring tools to analyse CPU, memory, I/O, and other key metrics.

***Identify your workload requirements**:* Once you have a good understanding of your database usage patterns, you need to identify your workload requirements. This includes understanding the number of concurrent users, the type of queries that are being run, and the amount of data being stored.

***Select the right instance type and size**:* AWS offers various instance types and sizes. The best instance type and size for your workload will depend on your specific requirements. Use the AWS RDS instance advisor tool to guide your selection of the right instance type and size.

***Monitor and adjust**:* Continuously monitor database performance after right-sizing it. This will help you to identify performance bottlenecks and make necessary adjustments to instance type and size.

Creating testable solutions with AWS Data Lab

Throughout the cloud adoption journey, challenges arise in database